

Project: Tripteron XYZ	Page 1/23
Calculation: finite element analysis in chassis	Date
Calculator: David Zuluaga Henao	Reviewer

Description

The chassis of the tripteron will be analyzed through finite element analysis (FEA) using the CAD/CAM software SolidWorks. The analysis will include stresses, buckling, displacement and deformation, considering the locations where the guides, lead screws, Nema motors, arms and effector are situated, which exert weight on the chassis beams. Consequently, this study will determine where the mentioned variables are concentrated and how they affect the stability and shape of the chassis.

Objective of the calculation

The objective of the finite element analysis is to find the values of the following variables:

- Shear stresses (MPa)
- Axial Stress (MPa)
- Bending stress (MPa)
- Buckling (AMPES)
- Displacement (mm)
- Strain

This in order to do a find if the stability of the chassis structure assures the correct functionality of the mechanism.

Model calculation variables

Variable	Description
F_1	Total weight 1
F_2	Total weight 2
F_3	Total weight 3
P_m	Weight of a nemu motor
P_t	Weight of a power screw
P_g	Weight of a guide
P_h	Weight of the arm
P_e	Weight of the effector

Table 1. Variables used

The variables used in the finite element analysis where the selected elements connected to the chassis applying a weight force in the beams. F_1 , F_2 and F_3 are the total weight applied to each of the beams which have an element in it.

Project: Tripteron XYZ	Page 2/23
Calculation: finite element analysis in chassis	Date
Calculator: David Zuluaga Henao	Reviewer

Image of the calculation model

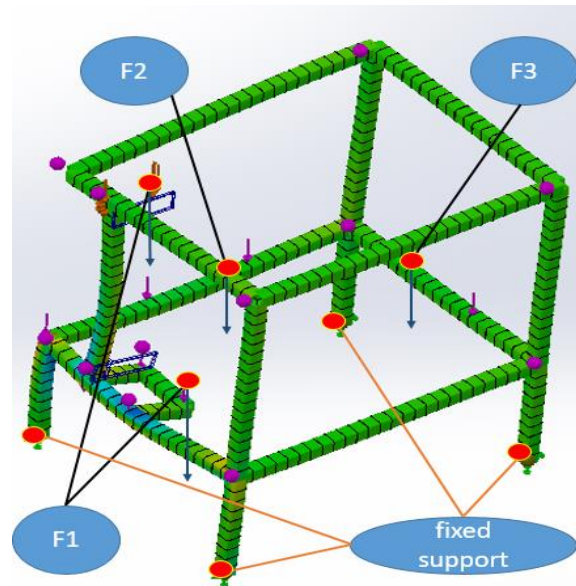


Figure 1. Diagram of applied forces and fixed support in Solidworks simulation

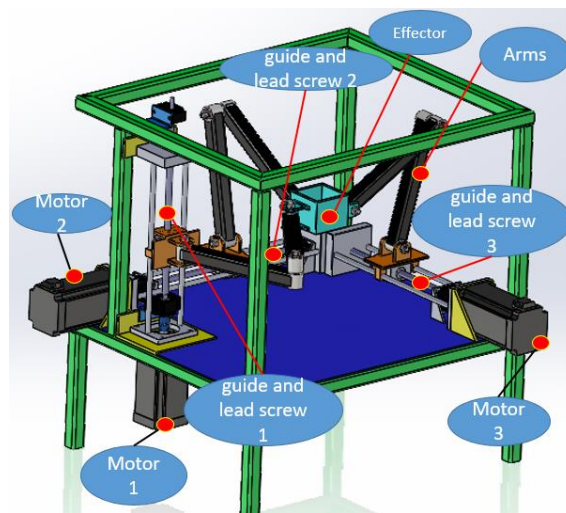


Figure 2. Diagram of elements applying weight on the chassis

As it can be seen in figure 1, the total weights were applied in each of the beams which the elements seen in figure 2 are on.

Project: Tripteron XYZ	Page 3/23
Calculation: finite element analysis in chassis	Date
Calculator: David Zuluaga Henao	Reviewer

Values of the calculation model

Variable	Description	Values
F_1	Total weight 1	9.762 Kg = 95.75 N
F_2	Total weight 2	9.762 Kg = 95.75 N
F_3	Total weight 3	9.762 Kg = 95.75 N
P_m	Weight of a nemu motor	5.216 Kg
P_t	Weight of a power screw	1.12 Kg
P_g	Weight of a guide	2.53 Kg
P_h	Weight of the arm	0.322 Kg
P_e	Weight of the effector	0.574 Kg

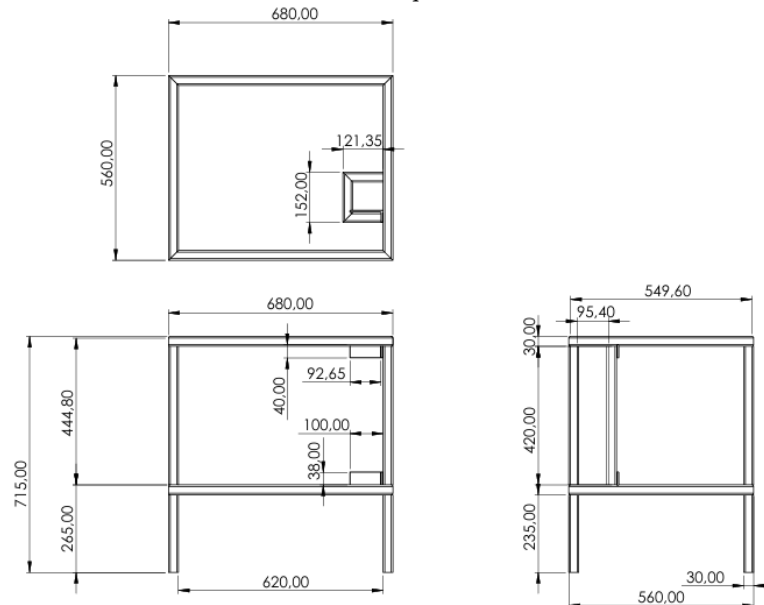
Table 2. data values

Each of the values in table 2 were taken using SolidWorks physical properties option, and the total weights were the sum of the elements which applied a weight to each beam. These weight values are approximate to the real ones, so they have a small error percentage.

Calculation model

For the calculation of the required variables, in the Program SolidWorks the following process was followed:

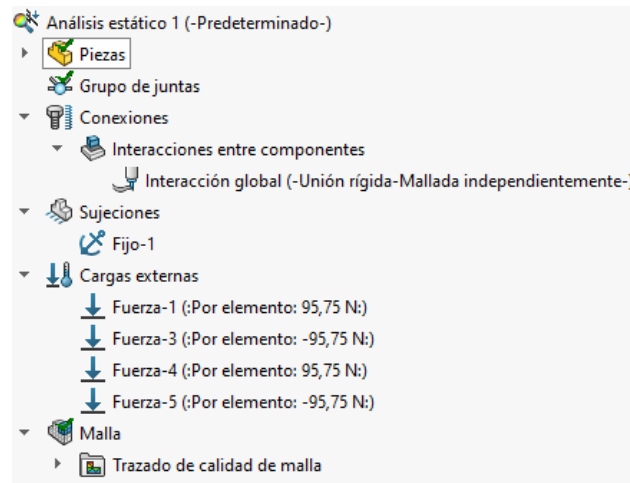
1. The CAD of the chassis was created with the required dimensions, seen in annex 1.



Annex 1. Chassis dimensions in mm.

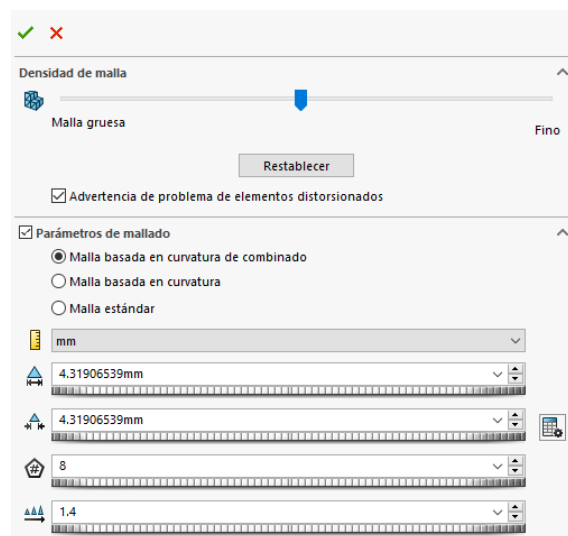
Project: Tripteron XYZ	Page 4/23
Calculation: finite element analysis in chassis	Date
Calculator: David Zuluaga Henao	Reviewer

- The material was defined as an **Alloy Steel** for the whole chassis.
- A static analysis was created with the following parameters:



Annex 2. Static analysis parameters

- As it can be seen in annex 1 the following parameters where used:
 - Component interaction: **Global interaction - rigid joint - independent meshing.**
 - Fixation: **Four fixed restraints.** The position can be seen in Figure 1.
 - External Loads: **Four forces of 95.75 N on their respective beam.** The position can be seen in Figure 1.
- The meshed definition was the following:

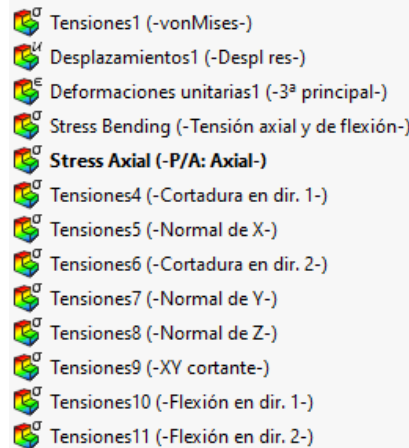


Annex 3. Mesh parameters

- As it can be seen in annex 2 the meshed parameter used was a **mesh based on combined curvature with a standard mesh density.**

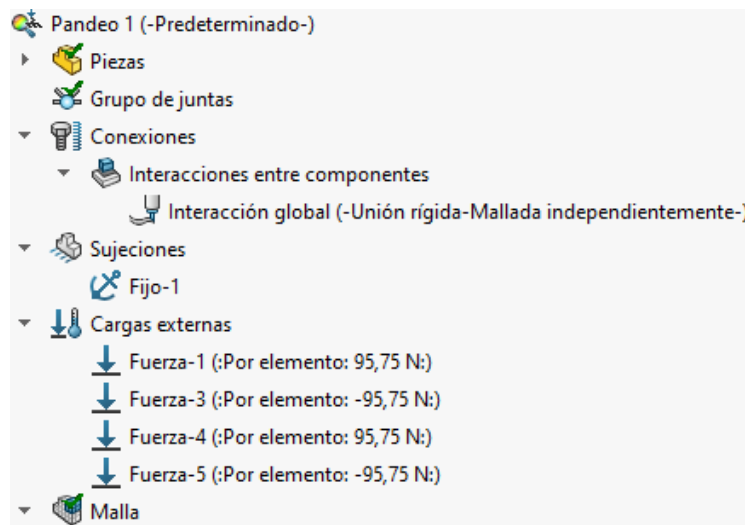
Project: Tripteron XYZ	Page 5/23
Calculation: finite element analysis in chassis	Date
Calculator: David Zuluaga Henao	Reviewer

6. The static analysis was executed and the studies were created as can be seen in Annex 3.



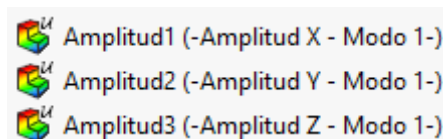
Annex 4. Static analysis studies

7. The Bucking analysis was created with the same parameters as the static analysis, as it can be seen in annex 4.



Annex 5. Buckling analysis parameters

8. The buckling analysis was executed and the studies were created as can be seen in Annex 5.



Annex 6. Buckling analysis studies

Project: Tripteron XYZ	Page 7/23
Calculation: finite element analysis in chassis	Date
Calculator: David Zuluaga Henao	Reviewer

Results and conclusions of the calculation model

The following results were taken from the static and buckling analysis as it can be seen in figure from 3 to 11.

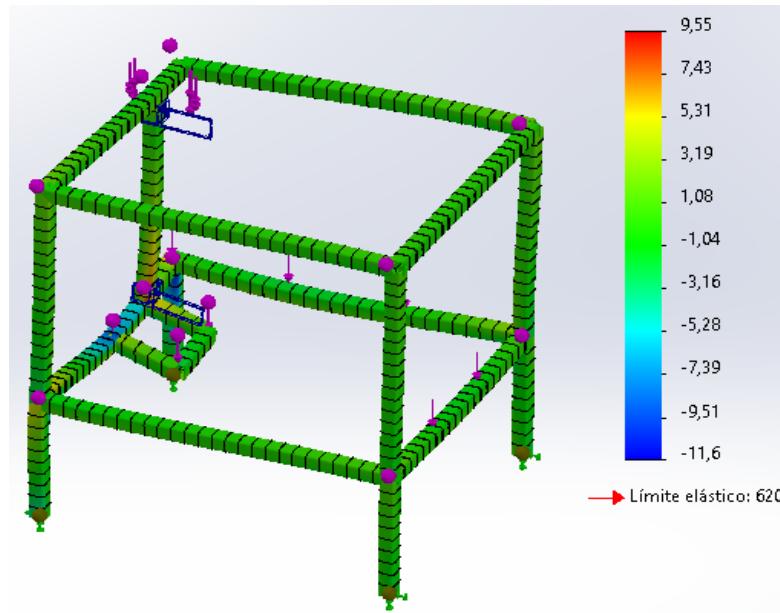


Figure 3. Bending Stress

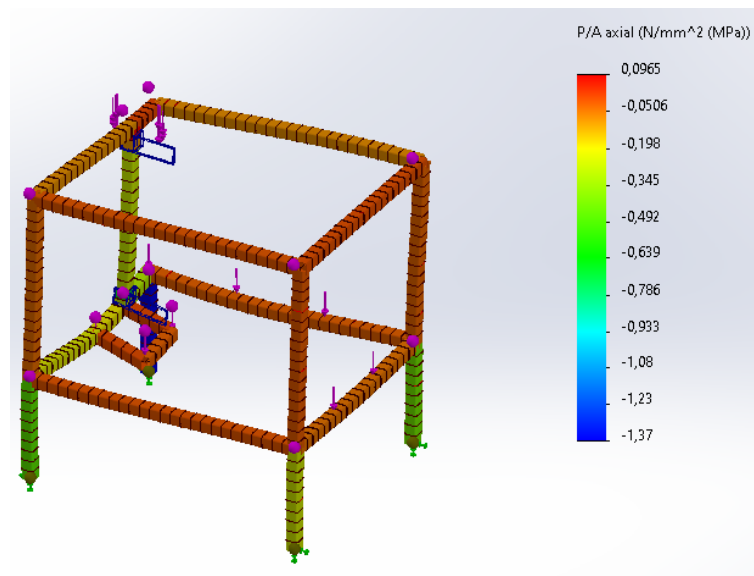


Figure 4. Axial Stress

Project: Tripteron XYZ	Page 8/23
Calculation: finite element analysis in chassis	Date
Calculator: David Zuluaga Henao	Reviewer

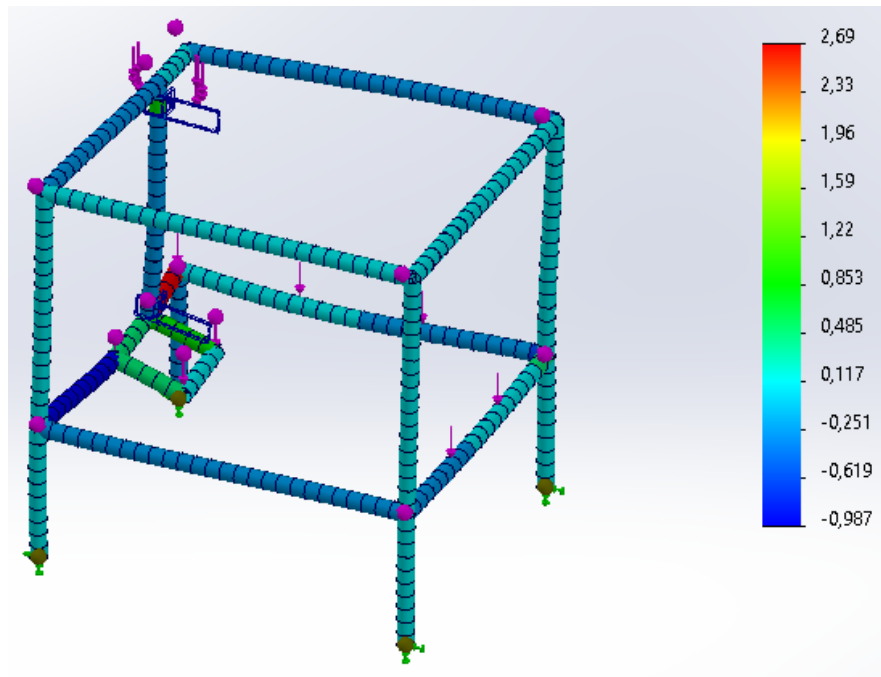


Figure 5. Shear stress dir 1. MPa

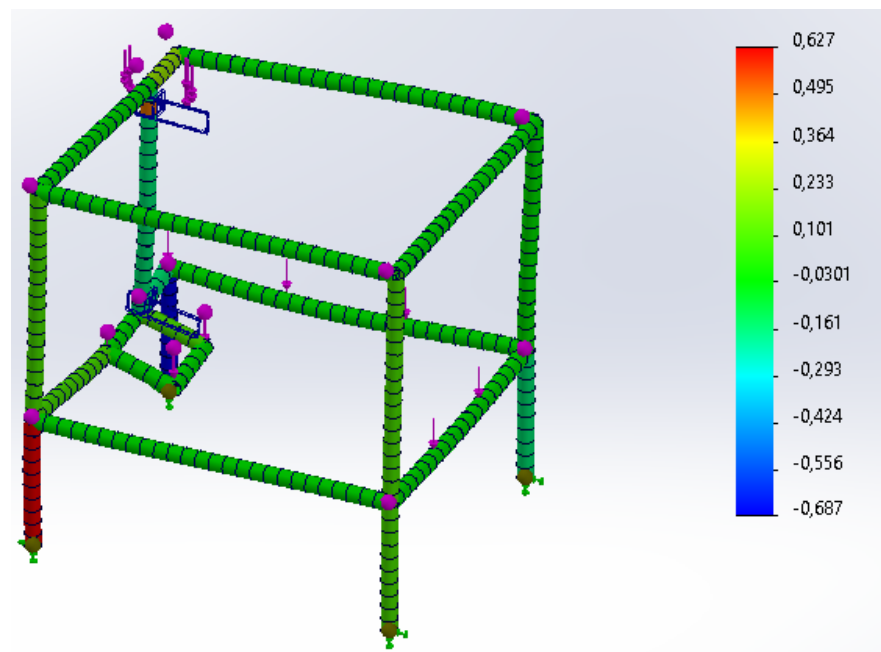


Figure 6. Shear stress dir 2. MPa

Project: Tripteron XYZ	Page 9/23
Calculation: finite element analysis in chassis	Date
Calculator: David Zuluaga Henao	Reviewer

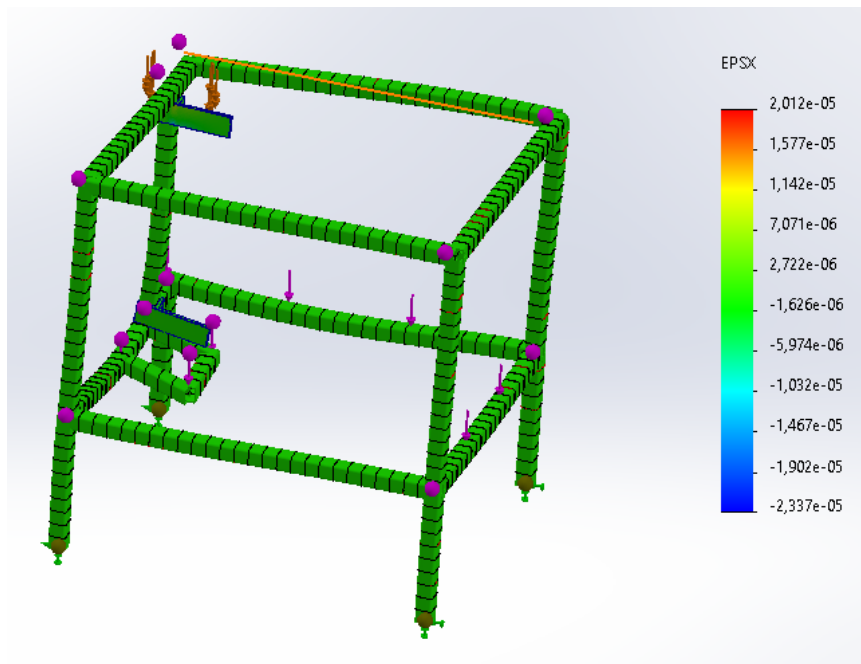


Figure 7. Strain

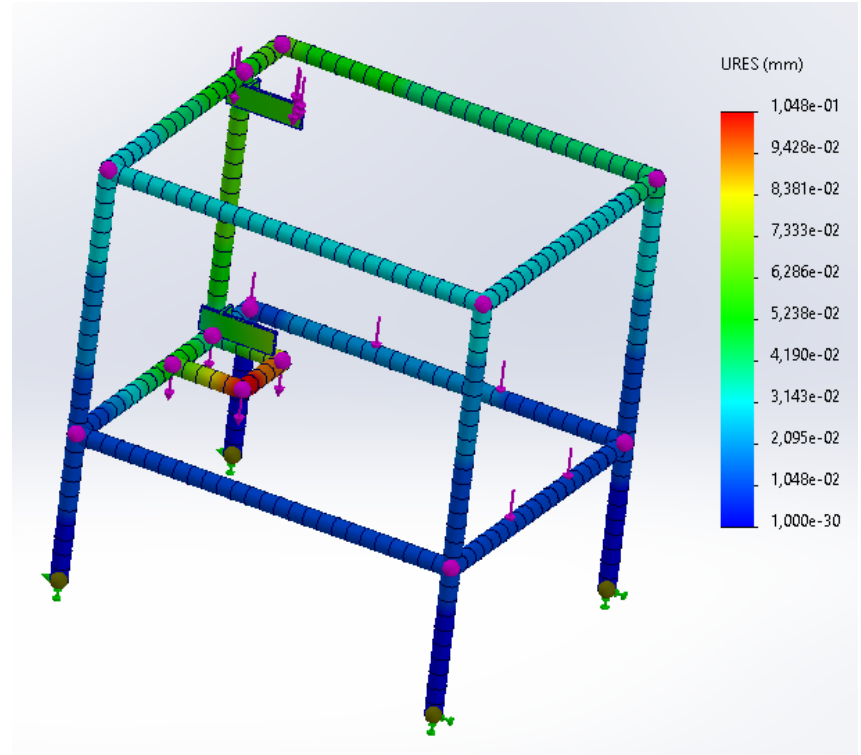


Figure 8. Displacement mm

Project: Tripterón XYZ	Page 10/23
Calculation: finite element analysis in chassis	Date
Calculator: David Zuluaga Henao	Reviewer

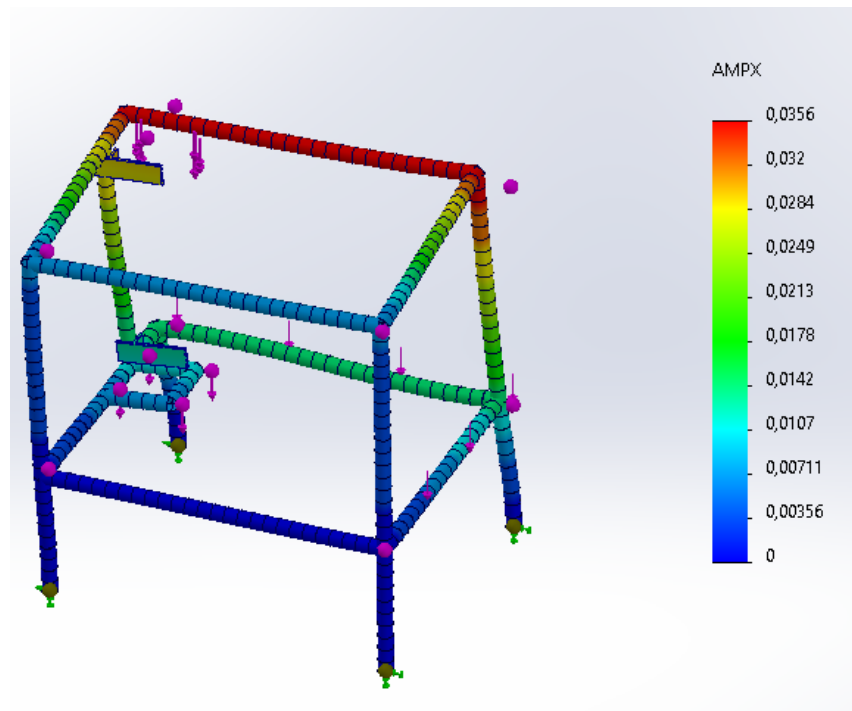


Figure 9. Buckling X

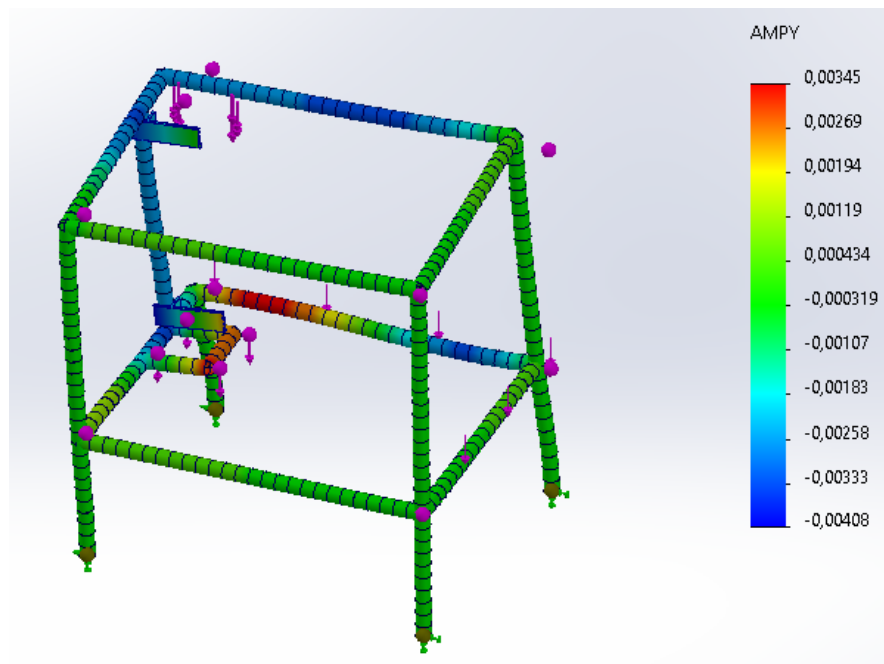


Figure 10. Buckling Y

Project: Tripteron XYZ	Page 11/23
Calculation: finite element analysis in chassis	Date
Calculator: David Zuluaga Henao	Reviewer

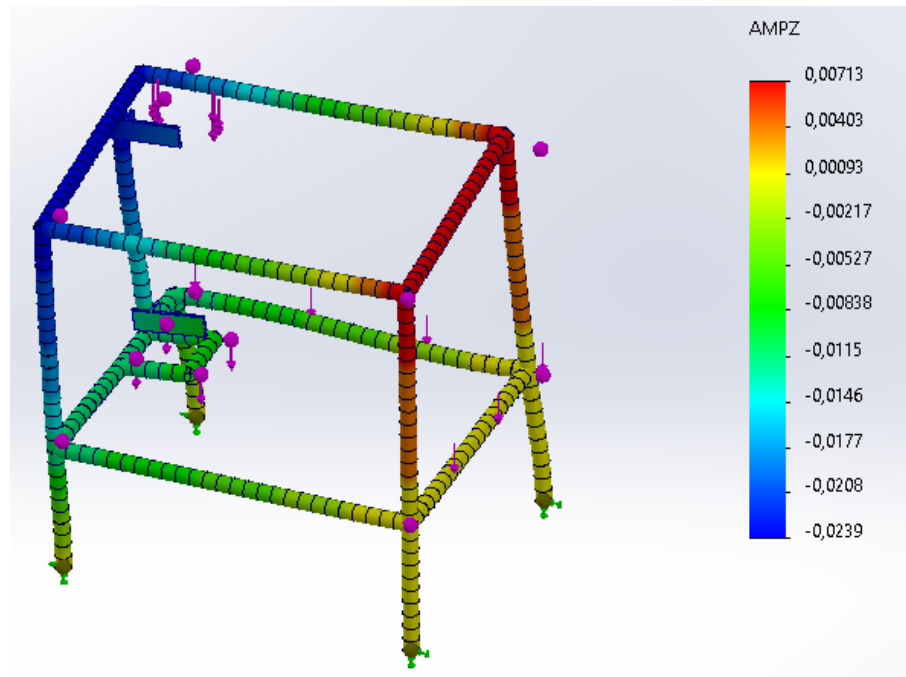


Figure 11. Buckling Z

Project: Tripteron XYZ	Page 12/23
Calculation: finite element analysis in chassis	Date
Calculator: David Zuluaga Henao	Reviewer

Graphic and value tables results

The following results were taken from the static and bucking analysis as it can be seen in figure from 12 to 20 and tables 3 to 12.

a) Bending Stress:

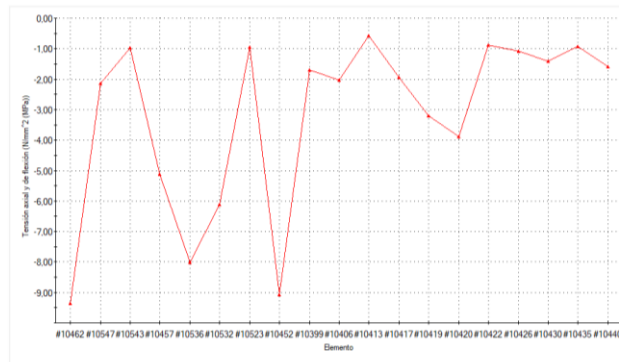


Figure 12. Bending stress graphic

Element	Bending Stress	X (mm)	Y (mm)	Z (mm)
10462	-9,36	568,739868	-481,862457	641,392334
10547	-2,14	447,389893	-481,862457	654,889221
10543	-0,982	447,389893	-481,862488	757,982239
10457	-5,13	568,739868	-481,862488	752,468262
10536	-8	568,739929	-449,689667	770,726563
10532	-6,11	568,739868	-340,029266	770,726563
10523	-0,952	568,739868	-121,434097	770,726563
10452	-9,08	568,739868	-481,862457	878,247559
10399	-1,71	458,346344	-481,862488	885,726563
10406	-2,04	290,388489	-481,862457	885,726501
10413	-0,574	108,752068	-481,862488	885,726563
10417	-1,95	8,75418091	-481,862488	885,726563
10419	-3,2	-39,6997376	-481,862488	885,726563
10420	-3,89	-64,7583618	-481,862488	885,726563
10422	-0,873	-81,2601013	-481,862488	843,804199
10426	-1,09	-81,2601013	-481,862488	745,603821
10430	-1,42	-81,2601013	-481,862488	650,824585
10435	-0,927	-81,2601013	-481,862488	525,719788
10440	-1,6	-81,2601013	-481,862457	387,177185

Table 3. Bending stress values

Project: Tripteron XYZ	Page 13/23
Calculation: finite element analysis in chassis	Date
Calculator: David Zuluaga Henao	Reviewer

b) Axial Stress:

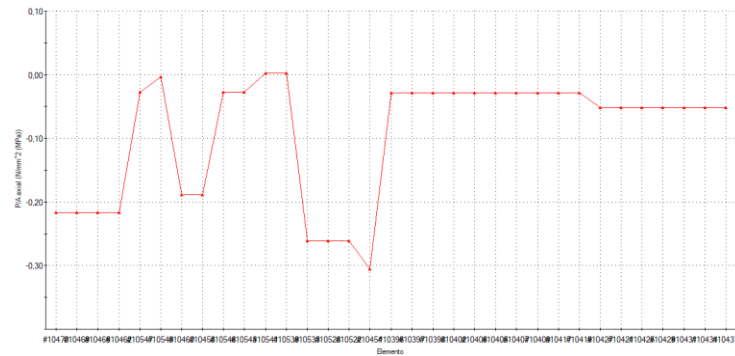


Figure 13. Axial stress graphic

Elemento	Axial Stress	X (mm)	Y (mm)	Z (mm)
10470	-0,216	568,73987	-481,86246	435,24399
10469	-0,216	568,73993	-481,86249	474,71512
10466	-0,216	568,73987	-481,86249	544,75916
10462	-0,216	568,73987	-481,86246	637,62976
10547	-0,0275	447,38989	-481,86249	665,49377
10549	-0,00286	537,64508	-481,86249	648,7265
10460	-0,189	568,73987	-481,86249	683,07983
10458	-0,189	568,73987	-481,86249	723,26263
10546	-0,0275	447,38992	-481,86249	689,42999
10543	-0,0275	447,38989	-481,86249	755,34192
10541	0,00286	473,5665	-481,86249	770,72656
10539	0,00286	525,30884	-481,86249	770,72656
10533	-0,261	568,73987	-359,26718	770,72656
10528	-0,261	568,73987	-239,12326	770,72656
10522	-0,261	568,73987	-91,967789	770,72656
10454	-0,305	568,73987	-481,86249	818,6405
10396	-0,0281	537,74811	-481,86249	885,72656
10397	-0,0281	508,21228	-481,86249	885,72656
10398	-0,0281	476,98828	-481,86249	885,72656
10402	-0,0281	375,75269	-481,86249	885,72656
10406	-0,0281	289,75073	-481,86246	885,7265
10405	-0,0281	296,95325	-481,86246	885,7265
10407	-0,0281	244,30492	-481,86246	885,7265
10409	-0,0281	204,92914	-481,86246	885,7265
10417	-0,0281	2,5223866	-481,86249	885,72656
10419	-0,0281	-42,389519	-481,86249	885,72656
10427	-0,0518	-81,260101	-481,86249	726,39679
10421	-0,0518	-81,260101	-481,86246	862,7417
10425	-0,0518	-81,260101	-481,86249	784,15674
10429	-0,0518	-81,260101	-481,86249	673,66138
10431	-0,0518	-81,260101	-481,86249	625,27557
10434	-0,0518	-81,260101	-481,86249	541,07648
10437	-0,0518	-81,260101	-481,86249	459,53357

Table 4. Axial stress values

Project: Tripterón XYZ	Page 14/23
Calculation: finite element analysis in chassis	Date
Calculator: David Zuluaga Henao	Reviewer

c) Shear stress dir 1.

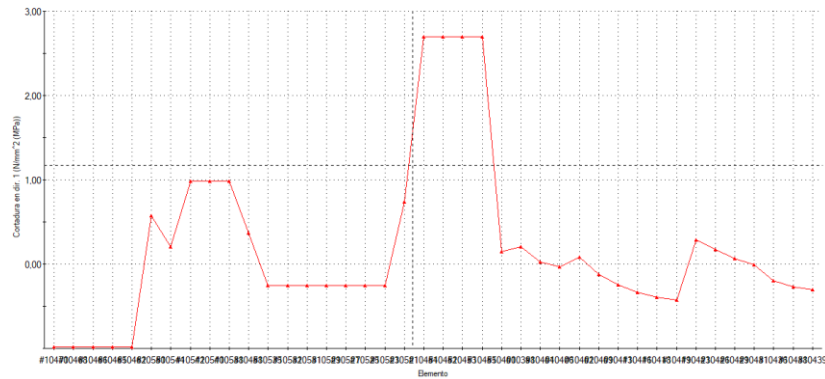


Figure 14. Shear stress dir 1. graphic

ShearStress1	X (mm)	Y (mm)	Z (mm)
-0,987	568,73993	-481,86249	531,17914
-0,987	568,73987	-481,86246	571,4729
-0,987	568,73987	-481,86249	625,53101
0,574	501,44727	-481,86249	648,72656
0,205	447,38992	-481,86249	731,90259
0,983	466,62238	-481,86249	770,72656
0,983	496,86169	-481,86249	770,72656
0,983	546,94592	-481,86249	770,72656
0,365	568,73987	-481,86249	739,01575
-0,256	568,73987	-173,97166	770,72656
-0,256	568,73987	-128,2354	770,72656
0,734	568,73987	-76,031181	770,72656
2,69	568,73987	-481,86249	819,24365
2,69	568,73987	-481,86249	810,40759
0,145	441,93991	-481,86249	885,72656
0,205	482,16956	-481,86249	885,72656
0,0252	325,47662	-481,86249	885,72656
-0,0347	286,56369	-481,86246	885,7265
0,085	372,81876	-481,86249	885,72656
-0,124	212,82373	-481,86249	885,72656
-0,244	106,94669	-481,86249	885,72656
-0,334	30,685913	-481,86249	885,72656
-0,394	-17,20911	-481,86249	885,72656
-0,424	-45,934914	-481,86249	885,72656
0,287	-81,260101	-481,86249	810,22418
0,176	-81,260101	-481,86249	741,29865
0,0645	-81,260101	-481,86249	666,90509
-0,00958	-81,260101	-481,86249	615,11597
-0,195	-81,260101	-481,86249	488,97107
-0,269	-81,260101	-481,86249	443,60199
-0,306	-81,260101	-481,86249	413,78043

Table 5. Shear stress dir 1. Values

Project: Tripteron XYZ	Page 15/23
Calculation: finite element analysis in chassis	Date
Calculator: David Zuluaga Henao	Reviewer

d) Shear stress dir 2.

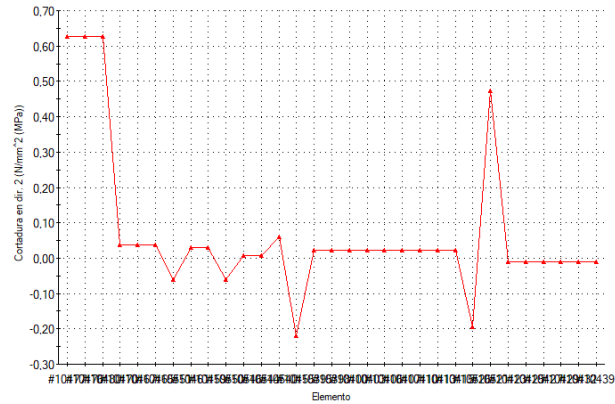


Figure 14. Shear stress dir 2. graphic

Shear Stress 2	X (mm)	Y (mm)	Z (mm)
0,627	568,73987	-570,20337	355,72653
0,627	568,73987	-541,49487	355,72653
0,627	568,73987	-651,40735	355,72653
0,0369	568,73987	-481,86249	442,76065
0,0369	568,73987	-481,86246	511,18112
0,0369	568,73987	-481,86249	556,79468
-0,0598	487,87149	-481,86249	648,72656
0,0307	568,73987	-481,86249	670,04102
0,0307	568,73987	-481,86249	721,76965
-0,0598	505,82135	-481,86249	648,72656
0,00621	447,38989	-481,86249	680,78986
0,00621	447,38992	-481,86249	733,12378
0,0598	509,42169	-481,86249	770,72656
-0,219	568,73987	-481,86249	812,84387
0,0233	552,72455	-481,86249	885,72656
0,0233	479,44061	-481,86249	885,72656
0,0233	440,23077	-481,86249	885,72656
0,0233	344,25281	-481,86249	885,72656
0,0233	288,41055	-481,86246	885,7265
0,0233	251,49352	-481,86246	885,7265
0,0233	180,22708	-481,86249	885,72656
0,0233	103,86248	-481,86249	885,72656
0,0233	51,560574	-481,86249	885,72656
-0,193	568,73987	-182,03851	770,72656
0,474	568,73993	-63,761681	770,72656
-0,0109	-81,260101	-481,86249	834,44385
-0,0109	-81,260101	-481,86249	762,70795
-0,0109	-81,260101	-481,86249	715,50659

Table 6. Shear stress dir 2. Values

Project: Tripteron XYZ	Page 16/23
Calculation: finite element analysis in chassis	Date
Calculator: David Zuluaga Henao	Reviewer

e) Strain

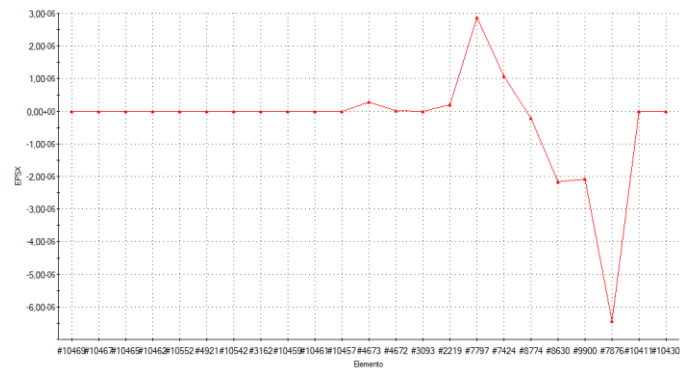


Figure 15. Strain graphic

Strain P1	X (mm)	Y (mm)	Z (mm)
0,00E+00	568,739868	-481,862488	633,55188
0,00E+00	457,18985	-481,862457	648,726501
4,59E-10	460,928162	-447,199829	749,376526
0,00E+00	455,165405	-481,862457	770,726563
2,09E-10	484,793884	-434,713684	749,376526
0,00E+00	568,739868	-481,862488	718,310486
0,00E+00	568,739868	-481,862488	670,67804
0,00E+00	568,739868	-481,862457	765,524902
2,84E-07	570,177185	-431,631165	749,376526
1,20E-08	508,772552	-442,067352	749,376526
-3,82E-09	510,210388	-464,526245	749,376526
2,01E-07	537,157837	-441,828888	749,376526
2,87E-06	549,643188	-58,2010651	749,376526
1,07E-06	527,937012	-58,7834511	749,376526
-2,13E-07	502,108978	-64,3615799	749,376526
-2,18E-06	541,639038	-71,3519211	749,376526
-2,09E-06	546,816589	-70,3942642	749,376526
-6,43E-06	556,256409	-76,1996765	749,376526
0,00E+00	158,614868	-481,862488	885,726563
0,00E+00	-81,2601013	-481,862488	637,468628

Table 7. Strain Values

Project: Tripteron XYZ	Page 17/23
Calculation: finite element analysis in chassis	Date
Calculator: David Zuluaga Henao	Reviewer

f) Displacement

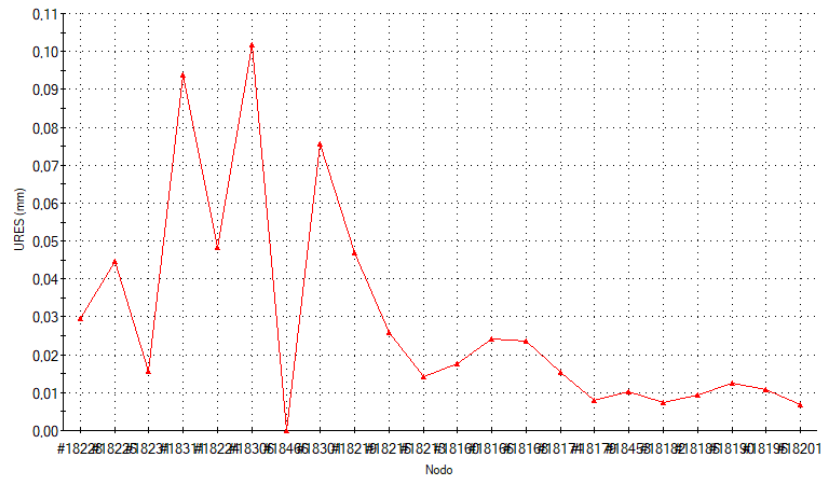


Figure 16. Displacement graphic

D (mm)	X (mm)	Y (mm)	Z (mm)
2,97E-02	568,739868	-481,862457	502,226532
4,46E-02	568,739868	-481,862457	575,476501
1,57E-02	568,739868	-481,862457	428,976532
9,37E-02	471,659912	-481,862488	648,726501
4,82E-02	568,739929	-481,862488	599,893188
1,02E-01	447,389893	-481,862488	673,126587
1,00E-30	447,389893	-464,398193	709,726501
7,56E-02	471,659912	-481,862488	770,726563
4,68E-02	568,739868	-481,862488	721,926575
2,59E-02	568,739868	-481,862488	816,726563
1,43E-02	568,739868	-481,862488	862,726563
1,77E-02	468,739899	-481,862488	885,726563
2,41E-02	318,739929	-481,862488	885,726563
2,36E-02	268,739899	-481,862488	885,726563
1,54E-02	118,739914	-481,862488	885,726563
7,84E-03	-6,26009607	-481,862488	885,726563
1,02E-02	-81,2601013	-406,862457	885,726501
7,25E-03	-81,2601013	-481,862457	860,488464
9,40E-03	-81,2601013	-481,862488	784,77417
1,26E-02	-81,2601013	-481,862488	658,583679
1,08E-02	-81,2601013	-481,862488	532,393188
6,89E-03	-81,2601013	-481,862457	380,96463

Table 8. Displacement Values

Project: Tripteron XYZ	Page 18/23
Calculation: finite element analysis in chassis	Date
Calculator: David Zuluaga Henao	Reviewer

g) Buckling X

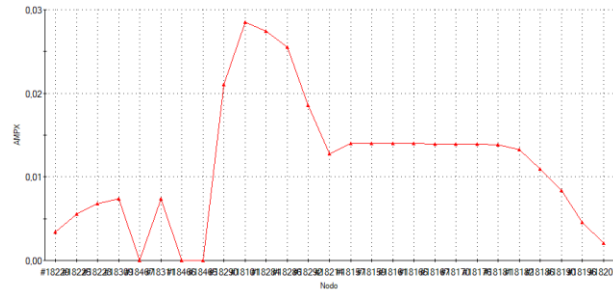


Figure 17. Buckling X graphic

Buckling X	X (mm)	Y (mm)	Z (mm)
0,00337	568,739929	-481,862488	477,809906
0,00555	568,739868	-481,862457	575,476501
0,00676	568,739868	-481,862488	624,309875
0,0074	520,19989	-481,862488	648,726563
0	508,064911	-465,556793	648,726501
0,0074	471,659912	-481,862488	648,726501
0	447,389893	-464,398193	709,726501
0	508,064911	-465,556793	770,726563
0,0211	568,739868	-281,862457	770,726563
0,0285	531,539917	-62,3440743	749,376526
0,0274	568,739868	-131,862457	770,726563
0,0255	568,739868	-181,862473	770,726563
0,0186	568,739868	-331,862488	770,726563
0,0128	568,739868	-481,862488	839,726563
0,014	543,739929	-481,862488	885,726563
0,014	493,739929	-481,862488	885,726563
0,014	443,739929	-481,862488	885,726563
0,014	343,739929	-481,862488	885,726563
0,014	293,739899	-481,862457	885,726501
0,0139	218,739914	-481,862488	885,726563
0,0139	68,7399139	-481,862488	885,726563
0,0138	-56,2600975	-481,862488	885,726563
0,0133	-81,2601013	-481,862457	860,488464
0,0109	-81,2601013	-481,862488	759,536072
0,00839	-81,2601013	-481,862488	658,583679
0,00454	-81,2601013	-481,862488	507,155121
0,00208	-81,2601013	-481,862488	406,202728

Table 9. Buckling X Values

Project: Tripteron XYZ	Page 19/23
Calculation: finite element analysis in chassis	Date
Calculator: David Zuluaga Henao	Reviewer

h) Buckling Y

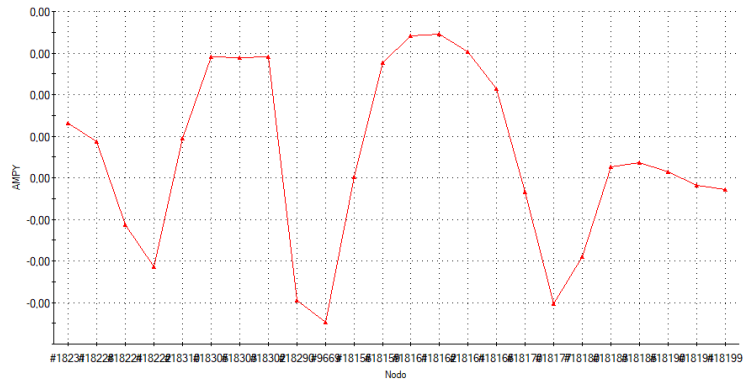


Figure 18. Buckling Y graphic

Buckling Y	X (mm)	Y (mm)	Z (mm)
0,00131	568,739868	-481,862457	428,976532
0,000879	568,739868	-481,862457	502,226532
-0,00114	568,739929	-481,862488	599,893188
-0,00215	568,739868	-481,862457	648,726501
0,000945	495,929932	-481,862488	648,726563
0,00291	447,389923	-481,862488	697,52655
0,00288	447,389893	-481,862488	746,326538
0,00292	447,389893	-481,862457	770,726563
-0,00295	568,739868	-281,862457	770,726563
-0,00348	578,999451	-62,1021881	749,376526
2,28E-05	568,739868	-481,862457	885,726501
0,00278	493,739929	-481,862488	885,726563
0,00342	443,739929	-481,862488	885,726563
0,00345	418,739929	-481,862488	885,726563
0,00303	368,739899	-481,862488	885,726563
0,00214	318,739929	-481,862488	885,726563
-0,000346	218,739914	-481,862488	885,726563
-0,00302	43,7398987	-481,862488	885,726563
-0,0019	-31,2601032	-481,862488	885,726563
0,000273	-81,2601013	-481,862488	835,250366
0,000356	-81,2601013	-481,862488	784,77417
0,000133	-81,2601013	-481,862488	658,583679
-0,000189	-81,2601013	-481,862488	557,631287
-0,000288	-81,2601013	-481,862488	431,440826

Table 10. Buckling Y Values

Project: Tripteron XYZ	Page 20/23
Calculation: finite element analysis in chassis	Date
Calculator: David Zuluaga Henao	Reviewer

i) Buckling Z

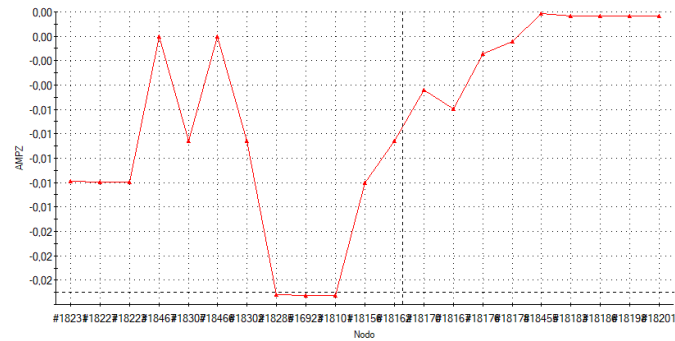


Figure 19. Buckling Z graphic

Buckling Z	X (mm)	Y (mm)	Z (mm)
-0,0119	568,739868	-481,862457	428,976532
-0,0119	568,739929	-481,862488	526,643188
-0,012	568,739868	-481,862488	624,309875
0	508,064911	-465,556793	648,726501
-0,00862	447,389893	-481,862457	648,726501
0	447,389893	-464,398193	709,726501
-0,00861	447,389893	-481,862457	770,726563
-0,0212	568,739868	-156,862473	770,726563
-0,0212	537,93988	-81,1021881	749,376526
-0,0213	531,539917	-62,3440743	749,376526
-0,012	568,739868	-481,862457	885,726501
-0,00856	418,739929	-481,862488	885,726563
-0,00442	218,739914	-481,862488	885,726563
-0,00593	293,739899	-481,862457	885,726501
-0,00142	68,7399139	-481,862488	885,726563
-0,000407	18,7399044	-481,862488	885,726563
0,00189	-81,2601013	-456,862488	885,726563
0,0017	-81,2601013	-481,862488	835,250366
0,0017	-81,2601013	-481,862488	759,536072
0,00169	-81,2601013	-481,862488	456,678925
0,00169	-81,2601013	-481,862457	380,96463

Table 11. Buckling Z Values

Project: Tripteron XYZ	Page 21/23
Calculation: finite element analysis in chassis	Date
Calculator: David Zuluaga Henao	Reviewer

Discussion

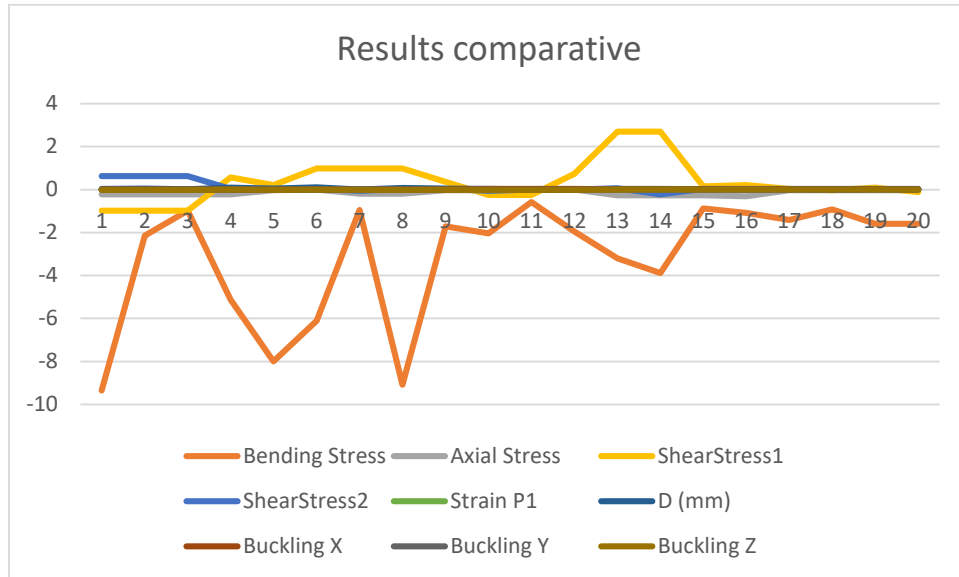


Figure 20. Results comparative graphic

	Bending Stress	Axial Stress	ShearStress1	ShearStress2	Strain P1	D (mm)	Buckling X	Buckling Y	Buckling Z
1	-9,36	-0,216	-0,987	0,627	0,00E+00	2,97E-02	0,00337	0,00131	-0,0119
2	-2,14	-0,216	-0,987	0,627	0,00E+00	4,46E-02	0,00555	0,000879	-0,0119
3	-0,982	-0,216	-0,987	0,627	4,59E-10	1,57E-02	0,00676	-0,00114	-0,012
4	-5,13	-0,216	0,574	0,0369	0,00E+00	9,37E-02	0,0074	-0,00215	0
5	-8	-0,0275	0,205	0,0369	2,09E-10	4,82E-02	0	0,000945	-0,00862
6	-6,11	-0,00286	0,983	0,0369	0,00E+00	1,02E-01	0,0074	0,00291	0
7	-0,952	-0,189	0,983	-0,0598	0,00E+00	1,00E-30	0	0,00288	-0,00861
8	-9,08	-0,189	0,983	0,0307	0,00E+00	7,56E-02	0	0,00292	-0,0212
9	-1,71	-0,0275	0,365	0,0307	2,84E-07	4,68E-02	0,0211	-0,00295	-0,0212
10	-2,04	-0,0275	-0,256	-0,0598	1,20E-08	2,59E-02	0,0285	-0,00348	-0,0213
11	-0,574	0,00286	-0,256	0,00621	-3,82E-09	1,43E-02	0,0274	2,28E-05	-0,012
12	-1,95	0,00286	0,734	0,00621	2,01E-07	1,77E-02	0,0255	0,00278	-0,00856
13	-3,2	-0,261	2,69	0,0598	2,87E-06	2,41E-02	0,0186	0,00342	-0,00442
14	-3,89	-0,261	2,69	-0,219	1,07E-06	2,36E-02	0,0128	0,00345	-0,00593
15	-0,873	-0,261	0,145	0,0233	-2,13E-07	1,54E-02	0,014	0,00303	-0,00142
16	-1,09	-0,305	0,205	0,0233	-2,18E-06	7,84E-03	0,014	0,00214	-0,000407
17	-1,42	-0,0281	0,0252	0,0233	-2,09E-06	1,02E-02	0,014	-0,000346	0,00189
17	-0,927	-0,0281	-0,0347	0,0233	-6,43E-06	7,25E-03	0,014	-0,00302	0,0017
18	-1,6	-0,0281	0,085	0,0233	0,00E+00	9,40E-03	0,014	-0,0019	0,0017
19	-1,6	-0,0281	-0,124	0,0233	0,00E+00	1,26E-02	0,0139	0,000273	0,00169

Table 12. Comparative values

Project: Tripteron XYZ	Page 22/23
Calculation: finite element analysis in chassis	Date
Calculator: David Zuluaga Henao	Reviewer

Taking into account figure 20 and table 12 which recompile the data of the figures and tables shown in the results, the following analysis was done by using the data collected:

Stresses Analysis

➤ **Bending Stress:**

- The bending stress values range from -9.36 to -0.574 MPa.
- The highest bending stress is observed in row 1 (-9.36 MPa), indicating a significant load or moment at this point, which could be a critical location for failure.

➤ **Axial Stress:**

- Axial stress values are relatively low, with a maximum of -0.305 MPa and a minimum of -0.216 MPa.
- These low axial stresses suggest that the primary loading condition is not purely axial but rather involves significant bending or shear components.

➤ **Shear Stresses:**

- Shear Stress 1 and Shear Stress 2 show variations, with Shear Stress 1 ranging from -0.987 to 0.983 MPa and Shear Stress 2 from 0.0369 to 0.627 MPa.
- These values indicate the presence of shear forces, which could contribute to potential shearing failures or deformations in the chassis.

Strain Analysis

➤ **Strain P1:**

- Strain values range from -3.82E-09 to 6.43E-06.
- These low strain values suggest that the material deformation is within the elastic range, implying that the chassis material is not undergoing plastic deformation under the applied loads.

Displacement Analysis

➤ **Displacement (D):**

- Displacements (D) range from 0 to 1.02E-01 mm.
- The maximum displacement observed is 0.102 mm, which is relatively small, indicating that the chassis maintains its structural integrity without significant deformations.

Project: Tripteron XYZ	Page 23/23
Calculation: finite element analysis in chassis	Date
Calculator: David Zuluaga Henao	Reviewer

Buckling Analysis

➤ **Buckling in X, Y, and Z Directions:**

- Buckling X values range from 0 to 0.0139.
- Buckling Y values range from -0.00346 to 0.00676.
- Buckling Z values range from -0.012 to 0.
- The buckling values are relatively small, with the highest observed buckling value being 0.0139 in the X direction.
- These values suggest that the chassis is stable under the given loading conditions, with no significant risk of buckling in any direction.

Conclusions

The FEA results for the tripteron machine chassis under various weight loads indicate that:

- The chassis experiences moderate bending stresses, with the highest being -9.36 MPa, which may require attention to ensure these stress levels do not exceed material limits.
- Axial stresses and strains are minimal, indicating that the primary loading conditions involve bending and shear forces.
- Shear stresses are present but within acceptable limits, suggesting that the chassis can handle shear forces without significant risk of shear failure.
- The displacements are minimal, ensuring that the structural integrity and alignment of the chassis are maintained.
- Buckling values are low, indicating that the chassis is stable and not prone to buckling under the applied loads.

Overall, the chassis design appears to be robust and capable of withstanding the applied loads with minimal deformation and a low risk of structural failure. However, attention should be given to areas with the highest bending stresses to ensure they are within safe limits, and further optimization may be considered if necessary.